Algorithms: Searching (original synthetic course)

Linear search checks elements in order. Worst-case comparisons: n.

Example: [4, 1, 7, 3], target 7. Check 4, then 1, then 7: 3 comparisons.

Binary search requires a sorted ascending array.

Use inclusive indices low=0, high=n-1. mid=(low+high)//2.

If a[mid]<target: low=mid+1; if larger: high=mid-1; else return mid.

Stop when low>high: target is absent. Worst-case time O(log n).

Example: [1,3,5,7,9], target 7. mid=2 (5), then mid=3 (7).